Introduction to Probability and Statistics

Session 4 Exercises

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Exercise 1: Chapter 9, problem 4 and 9

The following data indicate the relationship between x, the specific gravity of a wood sample, and Y, its maximum crushing strength in compression parallel to the grain

x_i	y_i(psi)	x_i	y_i(psi)
.41	1850	.39	1760
.46	2620	.41	2500
.44	2340	.44	2750
.47	2690	.43	2730
.42	2160	.44	3120

a) Plot a scatter diagram. Does a linear relationship seem reasonable?
b) Estimate the regression coefficients.
c) Predict the maximum crushing strength of a wood sample whose specific gravity is 0.43.
d) Estimate the variance σ^2 of an individual response.

Exercise 2: A measurement campaign involving two wireless devices was performed. One device (receiver) was kept on a fixed position, while the other (transmitter) was moved around and placed at several distances d_i for $i = 1, 2, \dots$ from the receiver. The transmitter was configured to send a known signal toward the receiver in every position and the receiver calculated the received power for every transmission. The file `received_power.csv` contains the $n = 500$ data obtained, in which P_i is the received power, measured in milliWatts [mW], and $d_i \in [30, 300]$ [m] is the distance between the transmitter and receiver.

a) Plot a scatter diagram. Does a linear relationship seem reasonable?

b) Use linear regression directly to estimate the regression coefficients. Then, compute the coefficient of determination, evaluate the standardized residuals and plot them. Can you describe the residual behavior?

c) A telecommunications engineer tells you that the relation between power and distance is

$$P_i = \frac{K}{d_i^{\gamma} L_i}$$

where K is a constant and L_i represents additional (random) losses. They also suggest to use the decibel [dB] version of the formula, so you should apply $10\log_{10}$ to each side of the equation before trying the linear regression. How the equation becomes? Can you identify which are the terms $Y_i, X_i, \beta_0, \beta_1, \epsilon_i$ in the new equations?

d) In the previous question, how should L_i be distributed in order to have ϵ_i satisfying all the assumptions for linear regression? (Tip: There is no need to evaluate the function of the distribution mathematically, just explain which distribution ϵ_i should have and try to discuss how L_i should be.)

e) Estimate the regression coefficients with the first $n' = 50$ measurements, compute the coefficient of determination, evaluate and plot the standardized residuals.

f) Estimate the regression coefficients with all the results $n = 500$, compute the coefficient of determination, evaluate and plot the standardized residuals. Comment the results comparing them with the ones obtained in points b) and e).

g) Give a possible estimate of K and γ based on the regression coefficients β_0 and β_1. Then, compute them numerically using the regression coefficients computed in e) and f),

h) Predict the mean response for new data $d_* = 400$ and compute the 95% prediction interval using the regression parameters estimated with $n' = 50$ and $n = 500$. Comment the results. (Tip: Remember that your x_* is not just d_*. You need to apply the transformation!)